MAT 362 HW11

Aditya Pahuja

Due 05/01

Problem. 4.5.1. Let $S \subseteq \mathbb{R}^3$ be a regular, compact, connected, orientable surface which is not homeomorphic to a sphere. Prove that there are points on S where the Gaussian curvature is positive, negative, and zero.

Solution. We first show that any compact surface has a point of positive curvature. Let $p \in S$ be a point of S whose distance from the origin is maximal (which exists by compactness). Then S lies entirely inside one of the two half-spaces demarcated by the plane $p + T_p(S)$, touching this tangent plane at exactly one point (notably p is normal to $T_p(S)$). This implies that the normal to any normal section at p will have the same direction as the vector p (and not the opposite direction), so both principal curvatures have the same sign, hence p is a point at which Gaussian curvature is positive. In fact, the Gaussian curvature is positive in a neighborhood of p because the curvature varies continuously. (More rigorously, one could apply a rotation and scale so that $p = (0, 0, 1)$ and $T_p(S)$ is the xy-plane, then see that S is the graph of a function with maximum 1 in a neighborhood of p.)

Since S is not homeomorphic to a sphere, its Euler-Poincaré characteristic is nonpositive, so the integral of K over S is nonpositive. The existence of an open set with positive Gaussian curvature at every point implies the existence of an open set with negative values of K (else the integral would be positive), and the intermediate value theorem shows us that there exists a point with curvature zero (say, by taking a path in S between a point with positive K and a point with negative K). $\square$

Problem. 4.5.2. Let T be a torus of revolution. Describe the image of the Gauss map of T and show, without using the Gauss-Bonnet theorem, that

$$\iint_T K \, d\sigma = 0.$$

Compute the Euler-Poincaré characteristic of T and check the above result with the Gauss-Bonnet theorem.

Solution. The image of the Gauss map of T is the entire sphere, since we can cover a point $(\cos\theta \sin\phi, \sin\theta \sin\phi, \cos\phi)$ on the sphere by taking the cross section given by rotating the generator of the torus in the xz-plane by θ and then moving an angle of ϕ within this cross section.

The characteristic of T is 0, as shown by the picture in the text. $\square$

Problem. 4.5.3. Let $S \subseteq \mathbb{R}^3$ be a regular compact surface with $K > 0$. Let $\Gamma \subseteq S$ be a simple closed geodesic in S, and let A and B be the regions of S which have Γ as a common boundary. Let $N\colon S \to S^2$ be the Gauss map of S. Prove that $N(A)$ and $N(B)$ have the same area.

Solution. Since the geodesic curvature of C is 0 everywhere and it is easily checked that the characteristic of A is 1, Gauss-Bonnet gives

$$\iint_A K \, d\sigma = 2\pi.$$

Then a change of variables gives

$$\iint_{N(A)} 1 \, d\overline{\sigma} = \iint_A K \, d\sigma = 2\pi$$

so $N(A)$ and $N(B)$ both have area 2π because the total surface area of S^2 is 4π. $\square$

Problem. 4.5.4. Compute the Euler-Poincaré characteristic of an ellipsoid.

Solution. Ellipsoids are diffeomorphic to spheres, so the characteristic of an ellipsoid is the same as that of a sphere, namely $\boxed{2}$. □

Problem. 4.5.5. Let C be a parallel of colatitude φ on an oriented unit sphere S^2, and let w_0 be a unit vector tangent to C at a point $p \in C$. Take the parallel transport of w_0 along C and show that its position, after a complete turn, makes an angle $\Delta\varphi = 2\pi(1 - \cos\varphi)$ with the initial position w_0. Check that

$$\lim_{R \to p} \frac{\Delta\varphi}{A} = 1 = \text{curvature of } S^2$$

where A is the area of the region R of S^2 bounded by C.

Solution. We have that the radius of C is $\sin\varphi$, so its circumference is $2\pi\sin\varphi$. Drawing the cone tangent to S^2 at C, the length from the apex of the cone to a point on C is $\tan\varphi$. If we "unroll" the cone into a sector of a circle with radius $\tan\varphi$ by cutting it along one of its generators, we can compute that the angle of the sector is $2\pi\cos\varphi$ by using the arc length and radius values; this is what gives us $\Delta\varphi = 2\pi - 2\pi\cos\theta$.

For the second part of the question we can calculate A with spherical coordinates to get

$$A = \int_0^{2\pi} \int_0^{\varphi} \sin\phi \; d\psi \; d\theta = 2\pi(1 - \cos\theta)$$

which gives the result...? □